# SafeSpec Automotive System Specification - v1.0

## Section 1: Exterior Lighting & Headlamp ECU Requirements

REQ-012: The headlamp shall switch to low beam when an oncoming vehicle is detected within 150 m.

REQ-017: The high beam shall activate above 30 km/h when ambient light is below 50 lux and no vehicle is ahead.

REQ-023: The headlamp shall react quickly to oncoming traffic.

REQ-025: Beam switching shall complete within 300 ms of the detection signal.

REQ-028: In case of sensor failure the headlamp should fall back to low beam.

REQ-031: The system shall enable high beam assist only above 40 km/h in darkness.

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## Section 2: Body Control & Diagnostic Requirements

REQ-034: The ECU shall turn off the headlamp output in daylight above 1000 lux.

REQ-041: The ECU shall log a DTC when the camera signal is lost for more than 500 ms.

REQ-050: When vehicle speed exceeds 20.0 km/h, the central door locking system shall lock all doors.

REQ-052: The ECU shall ignore speed signals flagged invalid.